OpenGeoSys Training 2026 – KIGAM/ZJU

Lecture 0: Course Overview | OGS Introduction

Olaf Kolditz, Mehran Ghasabeh, Haibing Shao, Xuerui Wang

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[2]Technische Universität Bergakademie Freiberg – TUBAF, Freiberg

[3]Technische Universität Dresden – TUD, Dresden

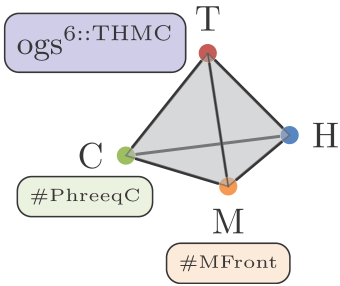
[4]Gesellschaft für Reaktorsicherheit – GRS, Braunschweig

Daejeon/Hangzhou, 14th October 2026

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Agenda

1. OGS Course Overview
2. OpenGeoSys Introduction
   - OGS-THMC
   - History
   - Software Engineering
   - OGSTools
   - Benchmark Gallery
   - ModelHub
3. References



14.10 OGS Introduction and installation (Haibing)

15.10 Hydromechanics - focus fractured rock (Olaf, Xuerui)

- OGS overview
- Continuum mechanics
- Hydromechanics
- Transport in fractured rock
- Exercises (afternoon session)
- Script available for lectures and exercises (github)
- (optional lecture on Digital Twins of Geosystems)

16.10 Geomechanics (Mehran)

17.10 Transport and reactive transport processes (Haibing)

today's course material:
`https://github.com/OlafKolditz/ogs-course-2026`

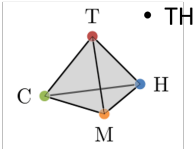
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Lectures

- Continuum Mechanics
- Hydromechanics
- Geomechanics

OGS Framework

- ogs^{THMC}
- ogs^{Tools}
- ogs^{Workflows}
- ogs^{DigitalTwins}



Exercises

- ogs installation
- H: Theis: Pumping test
- H: STIMTEC: Hydraulic tests
- H: Selke: Catchment
- M: Kirsch: Excavation
- HM: Fault reactivation
- THM: FE experiment

Part 1 morning session
lectures and software demonstration

Part 2 afternoon session (optional)
hands-on work with OGS (exercises)

Olaf

Lectures

- L0 Course Overview | OpenGeoSys Introduction
- L1 Continuum Mechanics
- L2 Hydromechanics
- L3 Groundwater Modeling

Overleaf: `https://www.overleaf.com/read/gpfdtpzqrjvp#ffc4c4`

Exercises

- E0 OpenGeoSys: OGS basics, Benchmark Gallery, GitHub
- E1 Theis: Pumping well
- E2 STIMTEC: Pumping test in fractured rock
- E3 Selke: Catchment modeling

GitHub: `https://github.com/OlafKolditz/ogs-course-2026`

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Mehran

Lectures

- L4 Underground Tunnel Excavation
- L6 Hydro-Mechanically Coupled Processes in Fault-Controlled Porous Medium
- L7 Heated Tunnel Problem

Exercises

- E0 Tunnel Convergence Problem
- E1 Theis: Pumping well
- E2 STIMTEC: Pumping test in fractured rock
- E3 Selke: Catchment modeling

github:
`https://github.com/mehranqsb/OGSTrainingCourse_KIGAM_ZJU`

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OpenGeoSys Introduction

OpenGeoSys THMC

Application areas



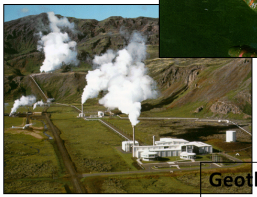
Water Resources





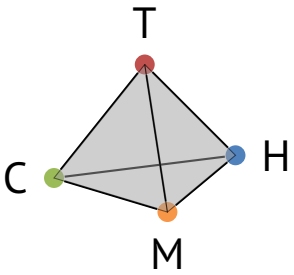
Water Quality

Geotechnical Systems





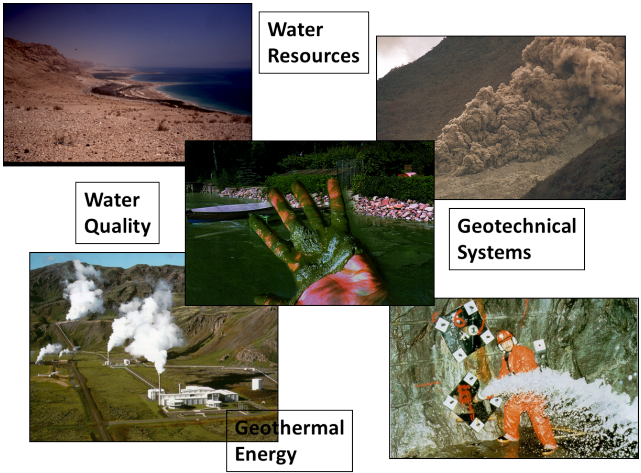
Geothermal Energy



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OpenGeoSys THMC

Framework



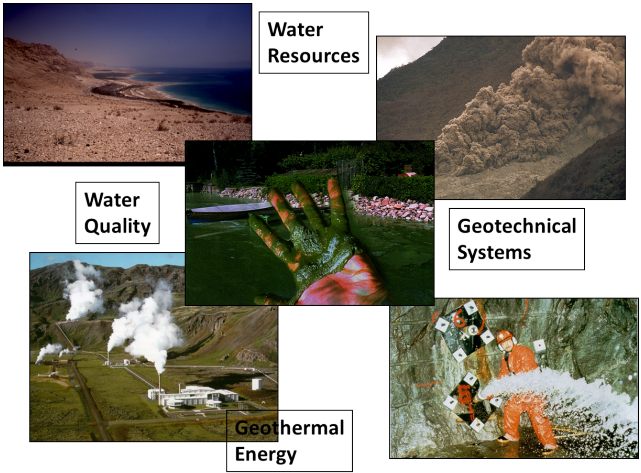
ogs-applications:

- T geothermal
- H geohydrological
- M geomechanical
- C geochemical (RTP)

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OpenGeoSys THMC

Framework



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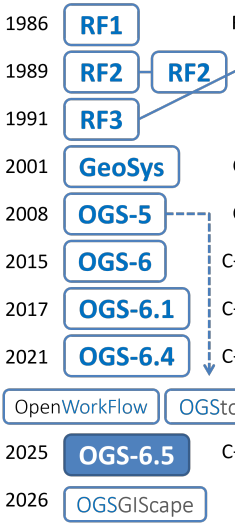
ogs-brands:

- ogs^{THMC}
- ogs^{Tools}
- ogs^{Workflows}
- ogs^{DigitalTwins}

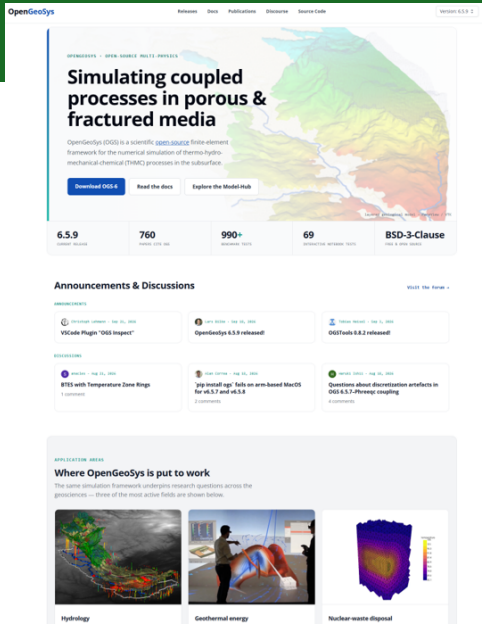
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OpenGeoSys

History | Website

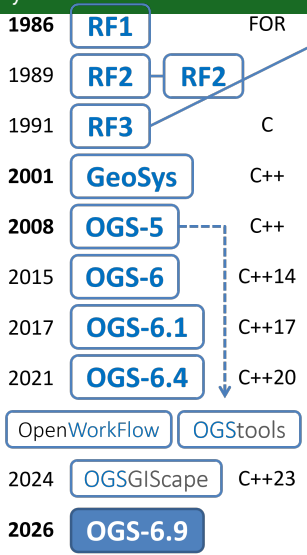


- *Dynamic data structures*
- *Object-orientation*
- *HPC*
- *Open source*
- *Software engineering*
- *Cross-platform*
- *DataExplorer*
- *Workflows*
- *Virtual Reality*
- *Python framework*
- *JupyterLab*
- *GIS based workflows*
- *Automation*
- *AI supported modeling*



OpenGeoSys

History



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- *Benchmark gallery*
- *Workflows (open)*
- *Automation*
- *AI supported development and modeling*
- *...*

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OpenGeoSys

History

1986 RF1 FOR

1989 RF2 RF2

1991 RF3 C

2001 GeoSys C++

2008 OGS-5 C++

2015 OGS-6 C++14

2017 OGS-6.1 C++17

2021 OGS-6.4 C++20

OpenWorkFlow OGStools

2024 OGSGIScape C++23

2026 OGS-6.9

FEFLOW

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OpenGeoSys

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OpenGeoSys

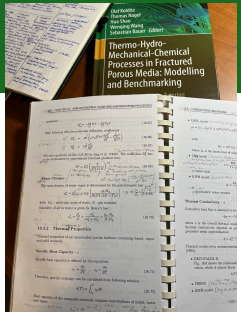
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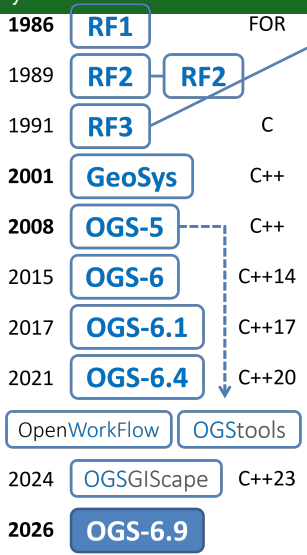




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OpenGeoSys

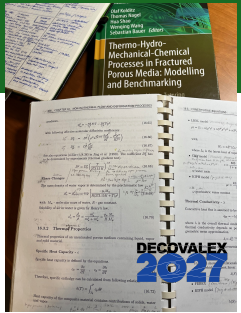
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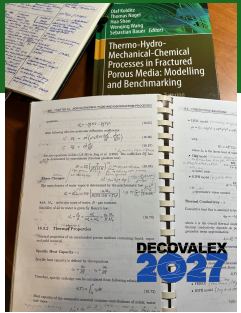
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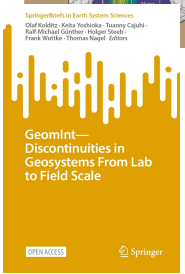
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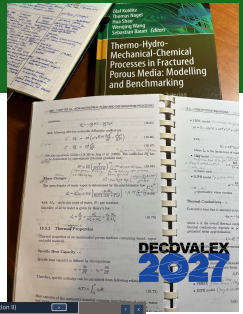
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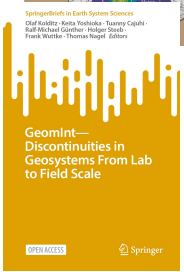
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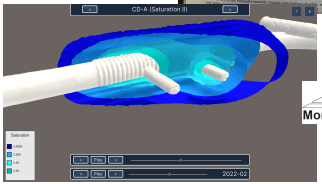
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OpenGeoSys

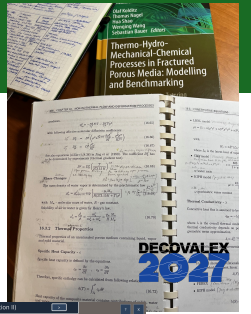
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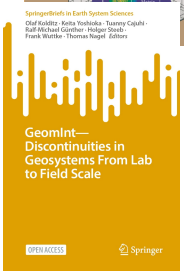
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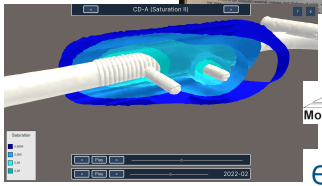
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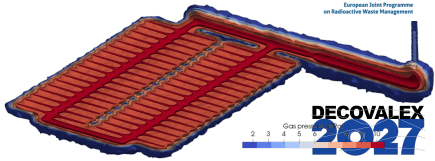












OpenGeoSys

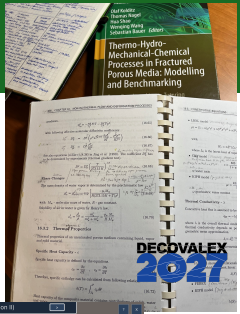
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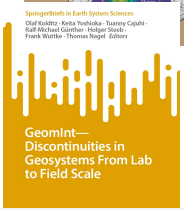
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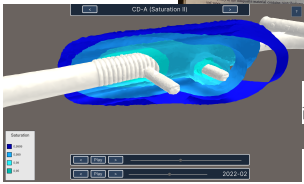
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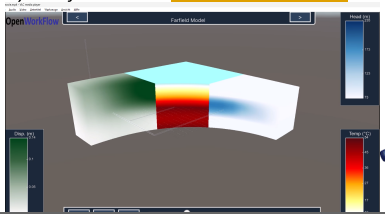


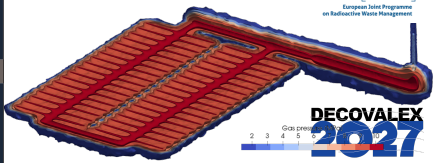


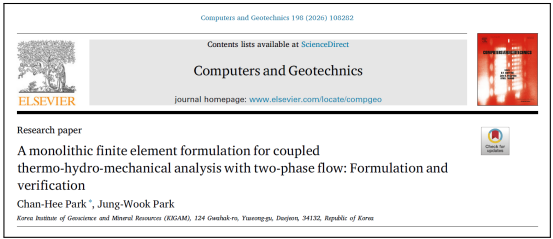












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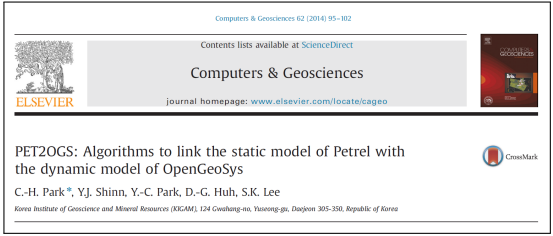
journal homepage: www.elsevier.com/locate/compgeo

Research paper

A monolithic finite element formulation for coupled thermo-hydro-mechanical analysis with two-phase flow: Formulation and verification

Chan-Hee Park *, Jung-Wook Park

Korea Institute of Geoscience and Mineral Resources (KIGAM), 124 Gwahak-ro, Yuseong-gu, Daejeon, 34132, Republic of Korea



Computers & Geosciences 62 (2014) 95–102

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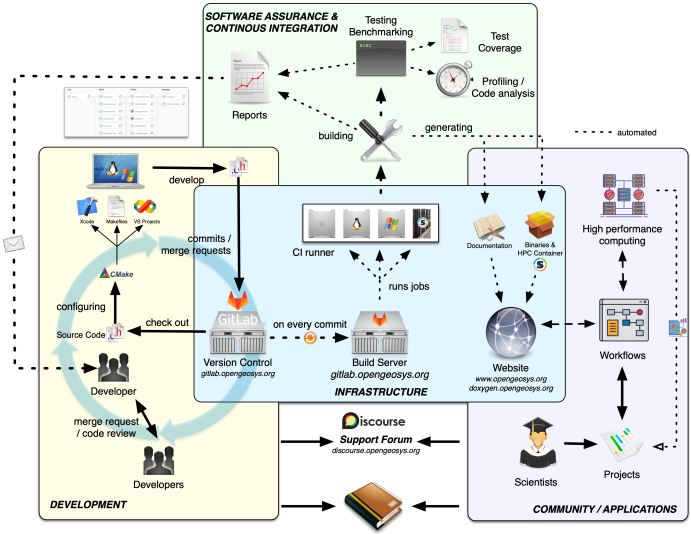
journal homepage: www.elsevier.com/locate/cageo

PET2OGS: Algorithms to link the static model of Petrel with the dynamic model of OpenGeoSys

C.-H. Park *, Y.J. Shinn, Y.-C. Park, D.-G. Huh, S.K. Lee

Korea Institute of Geoscience and Mineral Resources (KIGAM), 124 Gwahang-no, Yuseong-gu, Daejeon 305-350, Republic of Korea

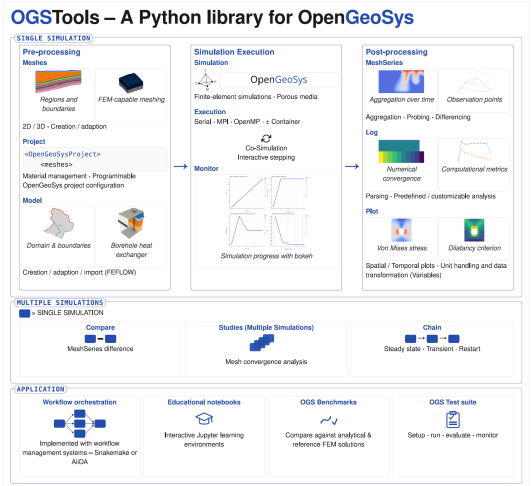
- PCH launched version in OGS 4/5 (github) and particle tracking
- OGS#PETREL interface for coupling structure and numerical models (C.-H. Park et al. 2014)
- OGS#FLAC/PFC hydro-mechanical coupling (J.-W. Park et al. 2025)
- monolithic TH2M model in OGS (Chan-Hee Park and J.-W. Park 2026)
- KIGAM is an important partner in DECOVALEX



Notes

- CI (continuous integration) system
- OGS chatbot (wip) for project files
- Guix for reproducibility
- AiiDA for provenance
- Data integration from DBs
- OGSTools for compact workflows

Ref: Bilke et al. (2025)

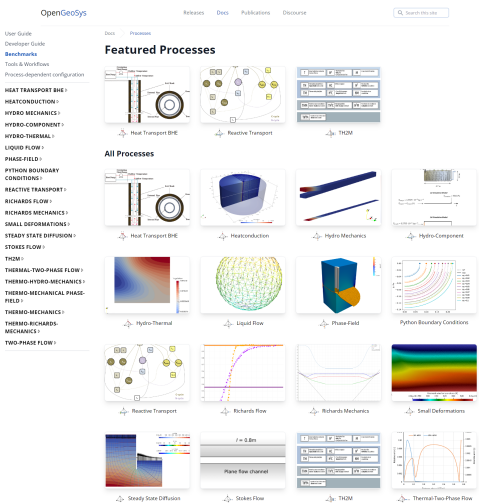


https://ogstools.opengeosys.org/stable/

Notes

- basics for building workflows
- comprehensive studies can be organized by OGSTools
  - automated grid convergence
  - ensemble Monte-Carlo runs for uncertainty studies
  - LLM assistant being developed
- OGSTools is the link to AI

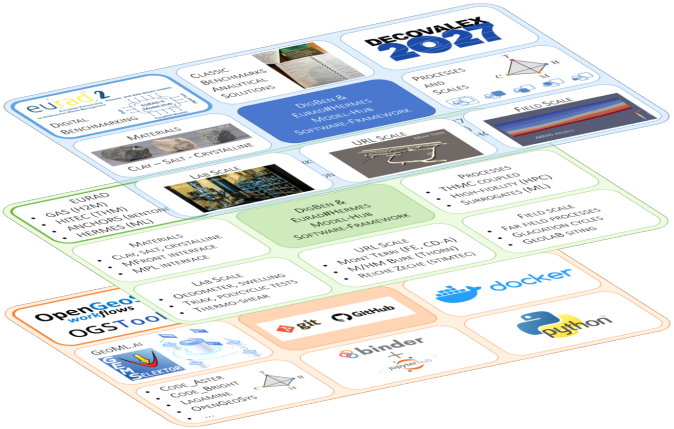
Ref: Meisel et al. (2026)



Notes

- Organized by processes (THMC concept)
- Starting point for own OGS models
- Jupyter notebooks (running on ogs-binder server)
- Context and complete workflows (relying on OGSTools)

https://www.opengeosys.org/6.5.9/docs/benchmarks/

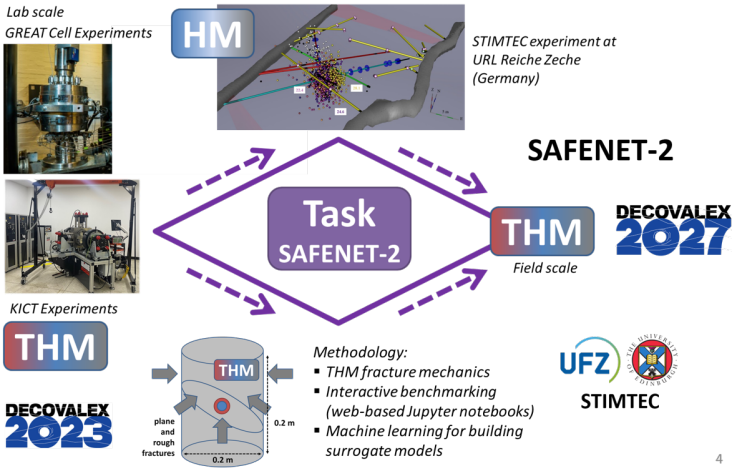


Notes

- Platform for joint benchmarking
- Initiated by EURAD and DECOVALEX projects
- Organized by application scales
- Interactive platform (via git)

Ref: Kolditz et al. (2026)

https://www.opengeosys.org/6.5.9/docs/hub/



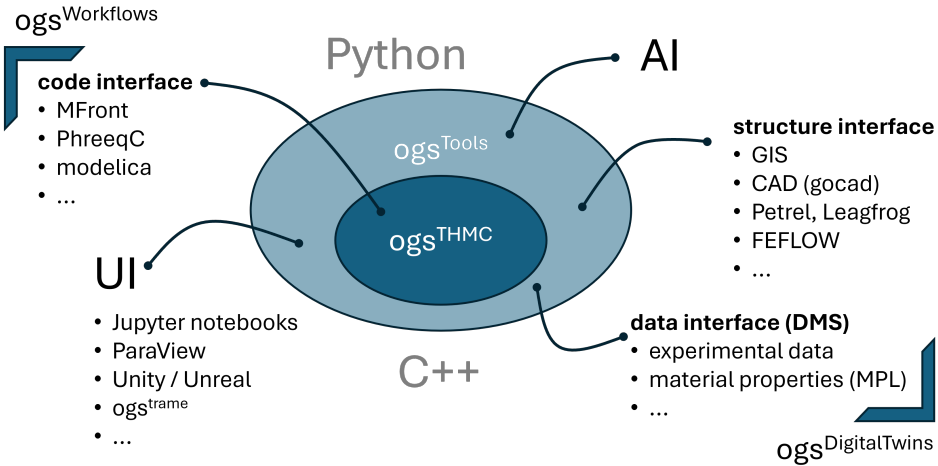
Korean cooperation partners:

- KIGAM
- SNU
- KAERI
- KICT (D2023)

STIMTEC experiment is part of the exercises

ZJU participation in next DECOVALEX-2031 ?

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ogsWorkflows
code interface
• MFront
• PhreeqC
• modelica
• ...
Python
AI
ogsTools
ogsTHMC
structure interface
• GIS
• CAD (gocad)
• Petrel, Leagfrog
• FEFLOW
• ...
UI
• Jupyter notebooks
• ParaView
• Unity / Unreal
• ogstrame
• ...
C++
data interface (DMS)
• experimental data
• material properties (MPL)
• ...
ogsDigitalTwins

How you can get involved - become part of the community:

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- providing interesting benchmarks (ideally as Jupyter notebooks) for the gallery

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- trying OGS for your studies and keep record in a repo (github, gitlab etc.)
- publish your studies together with code
- participating the annual community meeting

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Exercise
OGS Installation

OGS website (using Python installation package - pip)
`https://www.opengeosys.org/6.5.9/`
`https://www.opengeosys.org/6.5.9/docs/userguide/basics/5-mins-ogs/`

Haibing's lecture yesterday

Mehran's github:
`https://github.com/mehranqsb/OGSTrainingCourse_KIGAM_ZJU`

Olaf

Lectures

- L0 Course Overview | OpenGeoSys Introduction
- L1 Continuum Mechanics
- L2 Hydromechanics
- L3 Groundwater Modeling

Overleaf: `https://www.overleaf.com/read/gpfdtpzqrjvp#ffc4c4`

Exercises

- E0 OpenGeoSys: OGS basics, Benchmark Gallery, GitHub
- E1 Theis: Pumping well
- E2 STIMTEC: Pumping test in fractured rock
- E3 Selke: Catchment modeling

GitHub: `https://github.com/OlafKolditz/ogs-course-2026`

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[7] Chan-Hee Park and Jung-Wook Park. 'A monolithic finite element formulation for coupled thermo-hydro-mechanical analysis with two-phase flow: Formulation and verification'. In: Computers and Geotechnics 198 (2026). DOI: 10.1016/j.compgeo.2026.108282.

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